REPORT FOR PACMAN PROJECT (JAVA)

NAME: ALI RAZA

REG_ NO: 2025_CYS_40

1. Introduction

This project is a simple console-based Pacman game developed using Java. The objective of the game is to control Pacman inside a maze, collect food items, and avoid the ghost. The game demonstrates basic programming concepts and object-oriented principles.

2. Objectives

- To implement a simple game using Java
- To understand Object-Oriented Programming (OOP) concepts
- To practice user input handling and game logic
- To simulate a maze-based environment

3. Features of the Game

- *Maze-based game board (10×10 grid)*
- *Player-controlled Pacman*
- *Random movement of Ghost*
- *Food collection system*
- *Score tracking*
- *Game Over condition when ghost catches Pacman*

4. Technologies Used

- Programming Language: Java
- Concepts Used:

Classes and Objects

Encapsulation

Arrays (2D array for maze)

Loops and Conditions

Random class

User Input using Scanner

5. System Design

Main Classes:

1. GameBoard

- *Stores maze structure*
- *Displays game board*
- *Checks walls and food*

2. Pacman

- *Stores position (x, y)*
- *Tracks score*
- *Handles movement*

3. Ghost

- *Moves randomly*
- *Avoids walls and boundaries*

***4.* Food**

- *Represents food items*
- *Tracks whether food is eaten*

5. PacmanGame (Main Class)

- Controls game loop
- Handles user input
- Updates game state

6. Game Logic

- *Pacman starts from position (1,1)*
- *Ghost starts from position (8,8)*
- *Player moves using:*
    - *W (Up)*
    - *S (Down)*
    - *A (Left)*
    - *D (Right)*
- Pacman collects food (.) to increase score
- Ghost moves randomly in each iteration
- Game ends when Pacman collides with Ghost

7. How to Run the Program

1. *Compile the code:*

javac PacmanGame.java

2. *Run the program:*

java PacmanGame

8. Sample Output

- *Maze is displayed in console*
- *Pacman shown as P*
- *Ghost shown as G*
- *Walls shown as #*
- *Food shown as .*
- *Score updates after collecting food*

9. Conclusion

This project successfully demonstrates a simple implementation of a Pacman game using Java. It helps in understanding game logic, user interaction, and object-oriented design. The project can be further improved by adding graphics, levels, and multiple ghosts.

10. Future Improvements

- *Add GUI (using JavaFX or Swing)*
- *Multiple levels*
- *Better ghost AI*
- *Sound effects*
- *Timer-based gamepla*